

Terms

Coexistence. The set of challenges posed by the sharing of the spectrum by multiple communication technologies.

Colocation. A situation when more than one radio technology is used inside the same device. Colocated radios can sometimes interfere with each other.

on a single channel at a time.

Signal strength also has a huge impact. One is likely to experience more errors when signal strength is low (due to it being relatively close to background noise). The Bluetooth core specification states that a bit error rate (BER) of up to 0.1% is permitted. Anything beyond that is not acceptable.

The signal strength at which a radio can stay within this limit is known as its receiver sensitivity. So, if you are a product designer, choosing a radio with good receiver sensitivity is important.

Stack implementations often include several buffers for temporary packet storage. If the rate of packet arrival into a buffer exceeds the rate of packet consumption for a long enough period of time, the buffer will start to overflow and data will be lost.

Reliability challenges addressed by GFSK

The way in which radio signal properties are used for representing digital data is known as modulation scheme. The schemes based on frequency tend to be less susceptible to interference due to noise than those which are based on amplitude.

Bluetooth uses a special frequency based modulation scheme called Gaussian frequency shift keying (GFSK). It works by shifting a central carrier frequency upwards or downwards to represent either 1 or 0.

But abrupt changes in frequency can generate noise. Therefore, GFSK adds a filter, which causes frequency transitions to be smooth and less noisy. GFSK is a good fundamental platform on which to base your wireless communication.

Bluetooth packet fields

A Bluetooth packet has various fields. At the beginning is the preamble, which is 8-bit long and contains an alternating pattern of 1s and 0s. It is

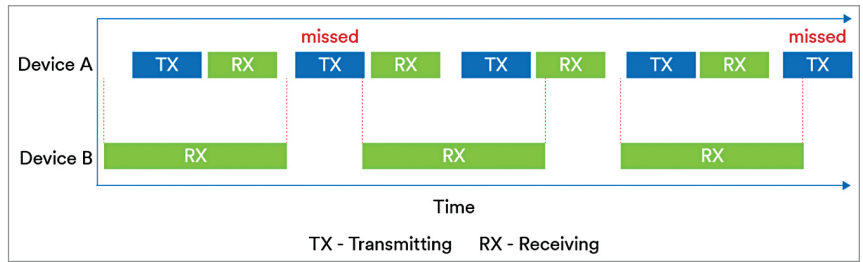


Fig. 2: Missed packets in unsynchronised communication

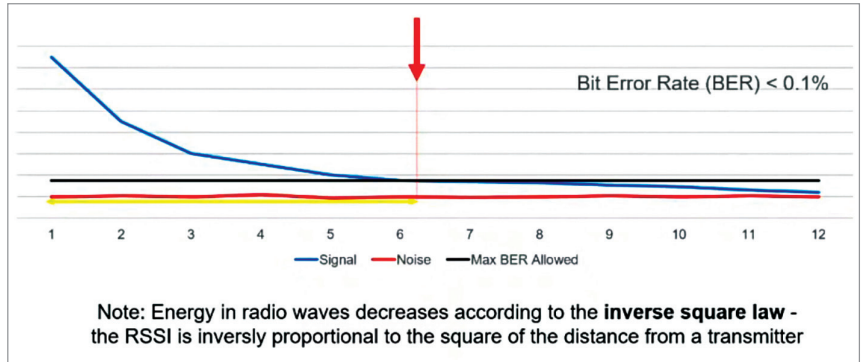


Fig. 3: Energy decrease with distance

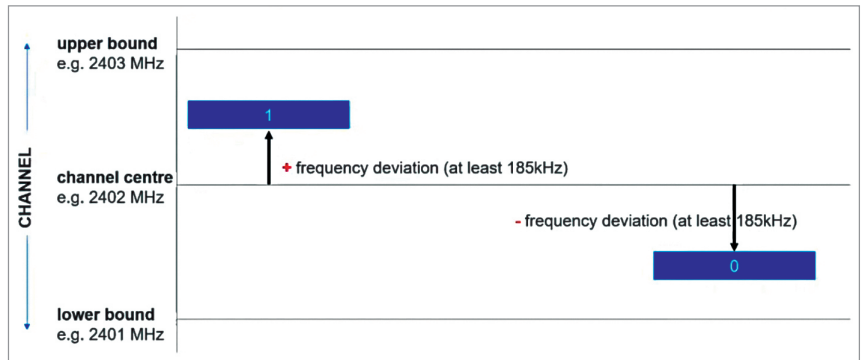


Fig. 4: GFSK shifts a central carrier frequency upwards or downwards to represent 1 or 0

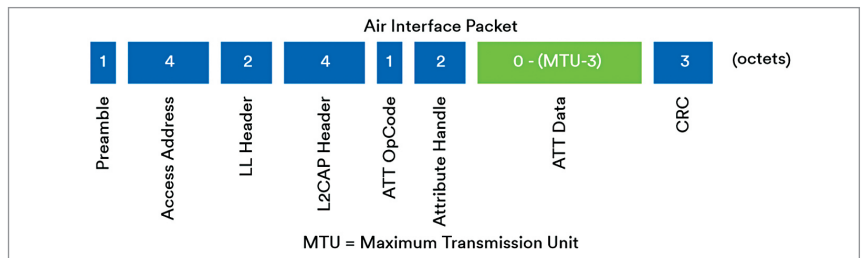


Fig. 5: The fields in a Bluetooth packet

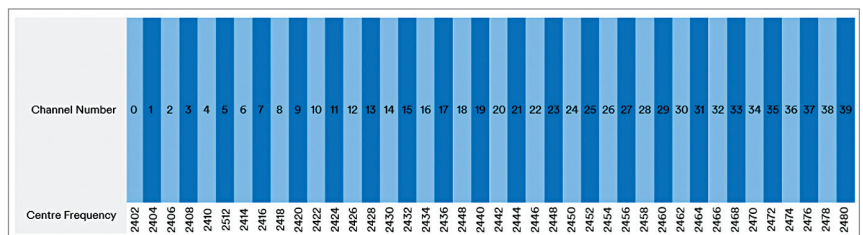


Fig. 6: 2.4GHz band divided into 40x2MHz channels